

2.3.17

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Question

Find the properties of the triangle with vertices $(4, 0)$, $(0, 3)$ and $(4, 0)$.

Theoretical Solution

Given the points,

Name	Point
A	$\begin{pmatrix} -4 \\ 0 \end{pmatrix}$
B	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Table: Variables used.

Theoretical Solution

We can calculate the magnitude of sides of the triangle using norms:

$$(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (1)$$

$$(\mathbf{B} - \mathbf{C}) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (2)$$

$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (3)$$

$$|\mathbf{A} - \mathbf{B}| = \sqrt{(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{B})} \quad (4)$$

$$|\mathbf{B} - \mathbf{C}| = \sqrt{(\mathbf{B} - \mathbf{C})^T \cdot (\mathbf{B} - \mathbf{C})} \quad (5)$$

$$|\mathbf{A} - \mathbf{C}| = \sqrt{(\mathbf{A} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{C})} \quad (6)$$

Calculation

$$(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -4 & -3 \end{pmatrix} \begin{pmatrix} -4 \\ -3 \end{pmatrix} \quad (7)$$

$$|\mathbf{A} - \mathbf{B}| = \sqrt{(-4)^2 + (-3)^2} \quad (8)$$

$$|\mathbf{A} - \mathbf{B}| = 5 \quad (9)$$

$$(\mathbf{B} - \mathbf{C})^T \cdot (\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -4 & 3 \end{pmatrix} \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (10)$$

$$|\mathbf{B} - \mathbf{C}| = \sqrt{(-4)^2 + (3)^2} \quad (11)$$

$$|\mathbf{B} - \mathbf{C}| = 5 \quad (12)$$

$$(13)$$

Theoretical solution

$$(\mathbf{A} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -8 & 0 \end{pmatrix} \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (14)$$

$$|\mathbf{A} - \mathbf{C}| = \sqrt{(-8)^2 + (0)^2} \quad (15)$$

$$|\mathbf{A} - \mathbf{C}| = 8 \quad (16)$$

$$\text{As } |\mathbf{A} - \mathbf{B}|^2 + |\mathbf{B} - \mathbf{C}|^2 = 25 + 25 = 50 \quad (17)$$

$$\text{and } |\mathbf{A} - \mathbf{C}|^2 = 64 \quad (18)$$

Hence, the triangle is **not RIGHT** angled.

Angles at the vertices A,B and C.

$$\cos(\angle A) = \frac{(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{C})}{|\mathbf{A} - \mathbf{B}| |\mathbf{A} - \mathbf{C}|} \quad (19)$$

$$\cos(\angle B) = \frac{(\mathbf{B} - \mathbf{A})^T \cdot (\mathbf{B} - \mathbf{C})}{|\mathbf{B} - \mathbf{A}| |\mathbf{B} - \mathbf{C}|} \quad (20)$$

$$\cos(\angle C) = \frac{(\mathbf{C} - \mathbf{A})^T \cdot (\mathbf{C} - \mathbf{B})}{|\mathbf{C} - \mathbf{A}| |\mathbf{C} - \mathbf{B}|} \quad (21)$$

$$(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 & 3 \end{pmatrix} \begin{pmatrix} -8 \\ 0 \end{pmatrix} = 32 \quad (22)$$

$$\cos(\angle A) = \frac{32}{5 \cdot 8} = 0.8 \quad (23)$$

$$\Rightarrow \angle A = \cos^{-1}(0.8) \approx 36.87^\circ \quad (24)$$

$$(\mathbf{B} - \mathbf{A})^T \cdot (\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -4 & -3 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = -16 + 9 = -7 \quad (25)$$

$$\cos(\angle B) = \frac{-7}{5 \cdot 5} = -0.28 \quad (26)$$

$$\Rightarrow \angle B = \cos^{-1}(-0.28) \approx 106.26^\circ \quad (27)$$

$$(\mathbf{C} - \mathbf{A})^T \cdot (\mathbf{C} - \mathbf{B}) = \begin{pmatrix} -8 & 0 \end{pmatrix} \begin{pmatrix} -4 \\ 3 \end{pmatrix} = 32 \quad (28)$$

$$\cos(\angle C) = \frac{32}{8 \cdot 5} = 0.8 \quad (29)$$

$$\Rightarrow \angle C = \cos^{-1}(0.8) \approx 36.87^\circ \quad (30)$$

$$\therefore \angle A \approx 36.87^\circ, \quad \angle B \approx 106.26^\circ, \quad \angle C \approx 36.87^\circ \quad (31)$$

$\therefore$ We can conclude that triangle **ABC** is an **ISOSCELES** triangle. 

```
#include <math.h>
#include <stdio.h>

// Function to compute side lengths and angles using matrix (
// vector dot product) method
void triangle_properties(double *sides, double *angles) {
    // Vertices: A(-4,0), B(0,3), C(4,0)
    double A[2] = {-4, 0};
    double B[2] = {0, 3};
    double C[2] = {4, 0};
```

```
// --- Compute side vectors ---  
double AB[2] = {B[0] - A[0], B[1] - A[1]};  
double BC[2] = {C[0] - B[0], C[1] - B[1]};  
double CA[2] = {A[0] - C[0], A[1] - C[1]};  
  
// --- Compute side lengths using dot product (matrix method)  
---  
sides[0] = sqrt(AB[0]*AB[0] + AB[1]*AB[1]); // AB  
sides[1] = sqrt(BC[0]*BC[0] + BC[1]*BC[1]); // BC  
sides[2] = sqrt(CA[0]*CA[0] + CA[1]*CA[1]); // CA
```

```
// --- Compute angles using dot product ---  
// A = angle between AB and AC  
double AC[2] = {C[0] - A[0], C[1] - A[1]};  
double dot_A = AB[0]*AC[0] + AB[1]*AC[1];  
double norm_AB = sqrt(AB[0]*AB[0] + AB[1]*AB[1]);  
double norm_AC = sqrt(AC[0]*AC[0] + AC[1]*AC[1]);  
angles[0] = acos(dot_A / (norm_AB * norm_AC)) * 180.0 / M_PI;  
  
// B = angle between BA and BC  
double BA[2] = {A[0] - B[0], A[1] - B[1]};  
double dot_B = BA[0]*BC[0] + BA[1]*BC[1];  
double norm_BA = sqrt(BA[0]*BA[0] + BA[1]*BA[1]);  
double norm_BC = sqrt(BC[0]*BC[0] + BC[1]*BC[1]);  
angles[1] = acos(dot_B / (norm_BA * norm_BC)) * 180.0 / M_PI;
```

```
// C = angle between CA and CB
double CB[2] = {B[0] - C[0], B[1] - C[1]};
double dot_C = CA[0]*CB[0] + CA[1]*CB[1];
double norm_CA = sqrt(CA[0]*CA[0] + CA[1]*CA[1]);
double norm_CB = sqrt(CB[0]*CB[0] + CB[1]*CB[1]);
angles[2] = acos(dot_C / (norm_CA * norm_CB)) * 180.0 / M_PI;
}
```

```
import matplotlib.pyplot as plt
import numpy as np
import math

# Define points
A = np.array([4, 0])
B = np.array([-4, 0])
C = np.array([0, 3])

# Function to compute angle at vertex 'v' between points 'a' and 'b'
def compute_angle(a, v, b):
    va = a - v
    vb = b - v
```

```
cos_theta = np.dot(va, vb) / (np.linalg.norm(va) * np.linalg.  
    norm(vb))  
cos_theta = np.clip(cos_theta, -1.0, 1.0)  
angle_rad = math.acos(cos_theta)  
return math.degrees(angle_rad)  
  
# Compute angles  
angle_A = compute_angle(B, A, C)  
angle_B = compute_angle(A, B, C)  
angle_C = compute_angle(A, C, B)  
  
# Create triangle path  
x_vals = [A[0], B[0], C[0], A[0]]  
y_vals = [A[1], B[1], C[1], A[1]]
```



```
# Plot triangle
plt.figure(figsize=(6, 6))
plt.plot(x_vals, y_vals, 'b-', linewidth=2)

# Plot points
plt.plot(A[0], A[1], 'ro')
plt.plot(B[0], B[1], 'go')
plt.plot(C[0], C[1], 'bo')
```

```
# Annotate points
plt.text(A[0]+0.2, A[1]-0.3, fA ({A[0]}, {A[1]})\nA  {angle_A:.2f
}, color='red')
plt.text(B[0]-2, B[1]-0.3, fB ({B[0]}, {B[1]})\nB  {angle_B:.2f},
color='green')
plt.text(C[0]-0.8, C[1]+0.2, fC ({C[0]}, {C[1]})\nC  {angle_C:.2f
}, color='blue')
```

```
# Axis settings
plt.axis('equal')
plt.grid(True)
plt.title(Triangle ABC with Angles)
plt.xlabel(X)
plt.ylabel(Y)

plt.savefig(../Figs/fig_3.png)
plt.show()
```

